DESCRIPTION OF COURSEWORK

Course Code	EEE406
Course Name	Power Systems Analysis
Lecturer	Dr Abdul Rashid Abdul Rahman
Academic Session	09/2025
Assessment Title	Laboratory 2 - Symmetrical and unsymmetrical fault analysis of a power system

A. Introduction/ Situation/ Background Information

Objectives

To become familiar with modelling and analysis of power systems under faulted condition and to compute the fault level, post-fault voltages and currents for different types of faults, both symmetrical and unsymmetrical.

Background Information

Symmetrical Components

In a three-phase system, a given set of unbalanced phasors may be replaced by three sets of balanced phasors which are referred to as the symmetrical components of the original unbalanced phasors.

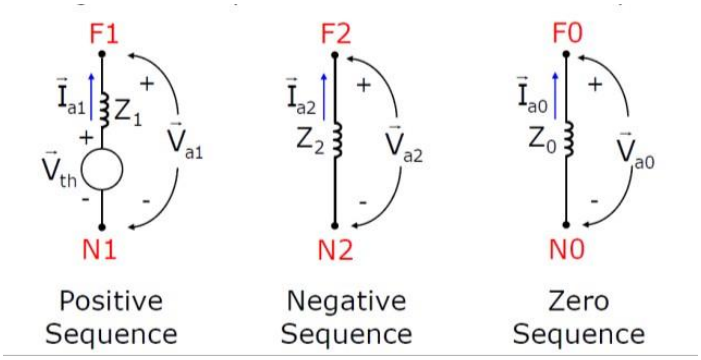
- The **positive-sequence** phasors are three-phase balanced and have the same phase sequence as the original set of unbalanced phasors.
- The **negative-sequence** phasors are three-phase balanced but having a phase sequence opposite to that of the original set of unbalanced phasors.
- The **zero-sequence** phasors are single-phase, equal in magnitude and in the same direction.

Sequence Networks

When the system is unbalanced, currents and voltages will contain positive-, negative-and zero-sequence components. The ratio of a sequence voltage to its corresponding sequence current represents a sequence impedance. Thus, the unbalanced power system can be replaced by three sequence networks.

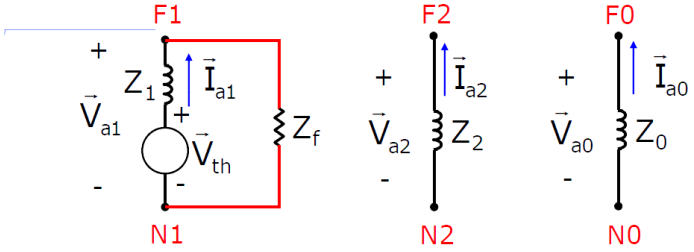
1. Positive-Sequence Network
2. Negative-Sequence Network
3. Zero-Sequence Network

The sequence network is a Thevenin equivalent representation of the power system between the fault point and neutral. It shows the sequence voltages and sequence currents at the fault point.



Symmetrical Fault:

- Balance three-phase fault



Sequence currents

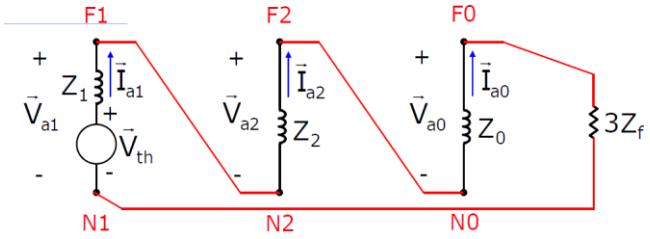
$$\vec{I}_{a1} = \frac{\vec{V}_{th}}{Z_1 + Z_f}$$

$$\vec{I}_{a0} = \vec{I}_{a2} = 0$$

  - Fault current $I_f = I_a^1$

Unsymmetrical Faults:

- Single line-to-ground Fault

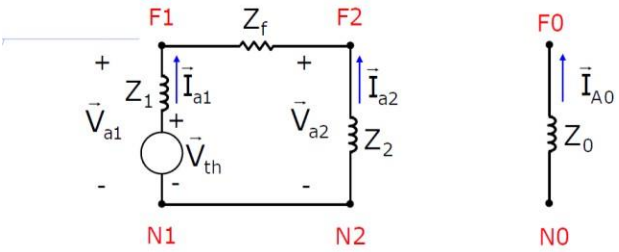


The sequence fault currents

$$\vec{I}_{a0} = \vec{I}_{a1} = \vec{I}_{a2} = \frac{\vec{V}_{th}}{Z_0 + Z_1 + Z_2 + 3Z_f}$$

  - Fault current $I_f = 3I_{a0}$

- Line-to-line Fault



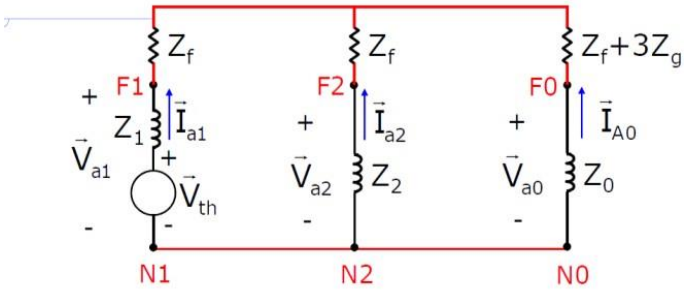
The sequence fault currents

$$\vec{I}_{a0} = 0$$

$$\vec{I}_{a1} = -\vec{I}_{a2} = \frac{\vec{V}_{th}}{Z_1 + Z_2 + Z_f}$$

- Fault current If = $-j\sqrt{3}\ I_a^1$

- Double Line-to-ground Fault



Let $Z_{0T} = Z_0 + Z_f + 3Z_g$

$Z_{1T} = Z_1 + Z_f$

$Z_{2T} = Z_2 + Z_f$

- Fault current If = $3I_a^o$

Pre-Laboratory

Knowledge on fault analysis and familiarity with PowerWorld Simulator software is very important prior to conducting this laboratory. Therefore, students should complete the Pre-Laboratory steps below:

1. Study the fault analysis through lecture notes or reference book.

2. Install the PowerWorld Simulator software.

3. Learn on how to use PowerWorld Simulator software from “PowerWorld Simulator Quick Guide Ver 5”.

Watch related tutorial video at powerworld.com or youtube.com websites for better understanding on using PowerWorld Simulator software.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 2 Evaluate faults in power system network.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
   ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
   iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.
2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

This is a group of **four (4) or five (5)** students's laboratory to be completed in 2 weeks. Simulation files(PowerWorld's Simulator pwb and pwd files) should be submitted with the report using filename

"Lab2ReportGroupX" in softcopy (PDF) for every task that should follow the following format:

1. Introduction
2. Methodology
3. Results and Discussion
4. Conclusion
5. References

Due Date: **26th December 2025 (Week 13).**

Software Tool: PowerWorld Simulator software.

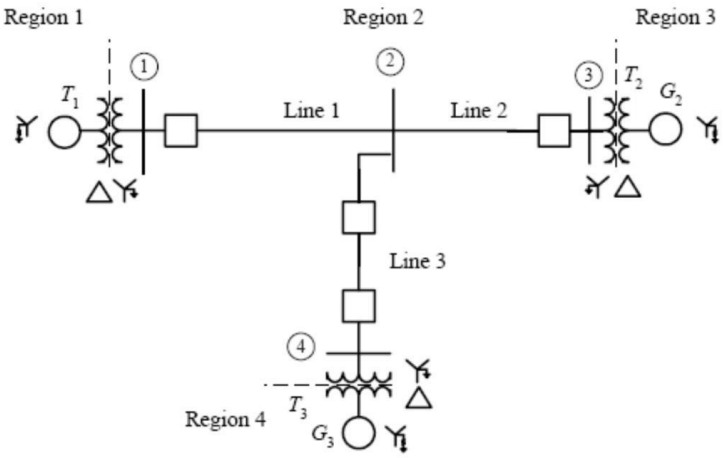
E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Design simple three-phase power system and determine fault currents and voltages for a single line-to-ground fault at bus 1 to bus 4.	30
2.	Perform manual calculation fault currents and voltages for a single line-to-ground fault at bus 1 to bus 4.	30
3.	Discuss on given questions in details.	30
4.	Completes report from introduction to conclusion	10
	TOTAL	**100**

F. Task(s)

A simple three-phase power system is shown below. Assume that the ratings of the various devices in this system are as follows:

- Generator *G*1: 250 MVA, 13.8 kV, R = 0.1 pu, X_1= 0.18 pu, X_2 = 0.15 pu, X_{G0} = 0.10 pu. This generator is grounded through an impedance $Z_N = j0.20$ pu.
- Generator *G*2: 500 MVA, 20.0 kV, R = 0.1 pu, X_1 = 0.15 pu, X_2 = 0.15, X_{G0} = 0.10 pu. This generator is grounded through an impedance $Z_N = j0.20$ pu.
- Generator G3: 250 MVA, 13.8 kV, R = 0.15 pu, X_1 = 0.20 pu, X_2 = 0.20, X_{G0} = 0.15 pu. This generator is grounded through an impedance $Z_N = j0.25$ pu.
- Transformer T1: 250 MVA, 13.8-Δ/240-Y kV, R = 0.01 pu, $X_1 = X_2 = X_0$ = 0.10 pu
- Transformer T2: 500 MVA, 20.0-Δ/240-Y kV, R = 0.01 pu, $X_1 = X_2 = X_0$ = 0.08 pu
- Transformer T3: 250 MVA, 13.8-Δ/240-Y kV, R = 0.01 pu, $X_1 = X_2 = X_0$ = 0.10 pu
- Each Line: R = 8 Ω, $X_1 = X_2$ = 40 Ω, X_0 = 80 Ω.



1. Assume prefault voltage of 1.05 per unit with Z_f = j0.10 pu, **determine** the fault currents and voltages for a single line-to-ground fault at bus 1 up to bus 4 by using **PowerWorld Simulator**.
2. Do a manual **calculation** for the fault currents and voltages for a single line-to-ground fault at bus 1 up to bus 4.
3. **Discuss** on the questions given below (must include reference in your discussion):
    a. What is the significance of the delta-Y connection of the transformers in this network?
    b. For ungrounded system, what will be the effect of ground faults?
4. Use the given table below as reference for your lab report.

Table 1 Fault Currents

Fault bus	Fault Current (p.u.)	Contributions to Fault Current				
		From bus	To bus	Phase A	Phase B	Phase C
1		2	1			
		4	2			
		3	2			
		G1	1			
		G2	3			
		G3	4			
2		1	2			
		3	2			
		4	2			
		G1	1			
		G2	3			
		G3	4			
3		2	3			
		1	2			
		4	2			
		G1	1			
		G2	3			
		G3	4			
4		1	2			
		2	4			
		3	2			
		G1	1			
		G2	3			
		G3	4			

Table 2 Bus Voltages

Fault bus	Bus	Bus Voltages during Fault (p.u.)		
		Phase A	Phase B	Phase C
1	1			
	2			
	3			
	4			
2	1			
	2			
	3			
	4			
3	1			
	2			
	3			
	4			
4	1			
	2			
	3			
	4			

APPENDIX 1

MARKING RUBRICS

Component Title	**Symmetrical and unsymmetrical fault analysis of a power system**				**Percentage (%)**		
Criteria	**Score and Descriptors**					**Weight (%)**	**Marks**
	Excellent (5)	**Good (4)**	**Average (3)**	**Need Improvement (2)**	**Poor (1)**		
System design and determination of fault currents and voltages.	**100% Complete and correct system network and fault currents and voltages for bus 1 to bus 4**	**80% Complete and correct system network and fault currents and voltages for bus 1 to bus 4**	**60% Complete and correct system network and fault currents and voltages for bus 1 to bus 4**	**40% Complete and correct system network and fault currents and voltages for bus 1 to bus 4**	**Uncomplete and Wrong system network and fault currents and voltages for bus 1 to bus 4**	**3**	
Manual calculation fault currents and voltages.	**100% detail steps of calculation shown with comprehensively and correctly evaluated**	**80% detail steps of calculation shown with comprehensively and correctly evaluated**	**60% detail steps of calculation shown with comprehensively and correctly evaluated**	**40% detail steps of calculation shown with comprehensively and correctly evaluated**	**less than 40% detail steps of calculation shown with comprehensively and correctly evaluated**	**3**	
Discuss on given questions in details.	**100% detail discussion and explanation on given questions**	**80% detail discussion and explanation on given questions**	**60% detail discussion and explanation on given questions**	**40% detail discussion and explanation on given questions**	**less than 40% detail discussion and explanation on given questions**	**3**	
Completes report from introduction to conclusion	**100% complete laboratory report with all details as highlighted in the task.**	**80% complete laboratory report with all details as highlighted in the task.**	**60% complete laboratory report with all details as highlighted in the task.**	**40% complete laboratory report with all details as highlighted in the task.**	**less than 40% complete laboratory report with all details as highlighted in the task.**	**1**	
					TOTAL		

Note to students: Please include the marking rubric when submitting your coursework.